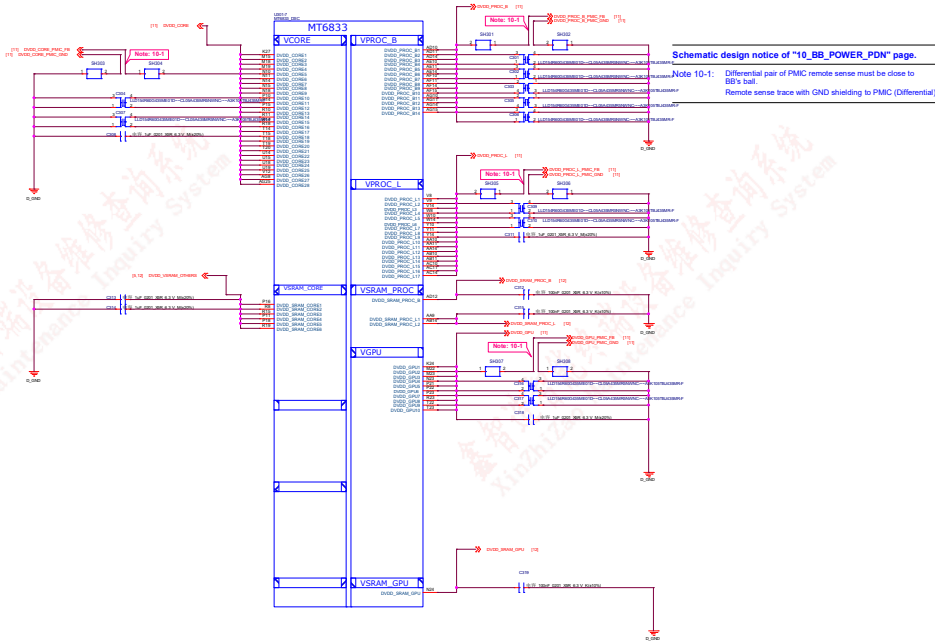
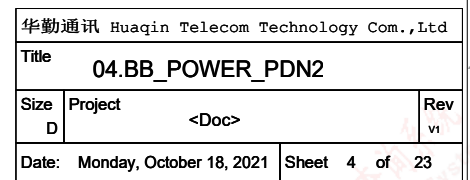


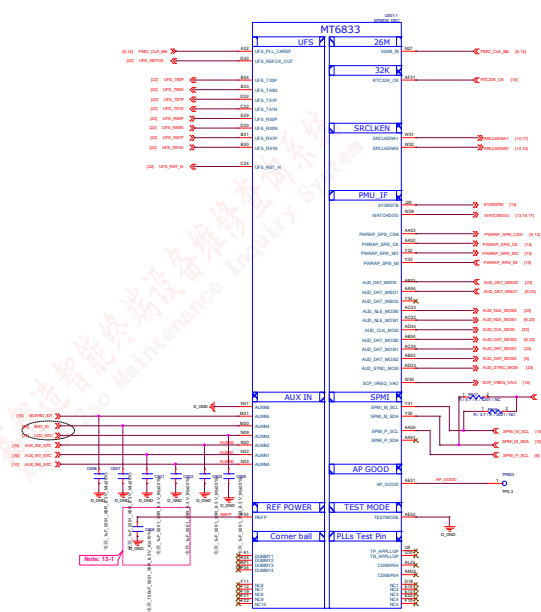
Date	Version	Page	Change notice
2020.06.xx	Pre-Released		
2020.07.xx	V01		
2020.08.14	V02	13_BB_1 29_POWER_EXT 20_POWER_MT6365_Buck 25_POWER_MT6315_VMD	Disconnect Dummy pins A1,A34,AP1,AP34 to GND Change C2923 cap from 6.3V to 10V [PDN] Change Caps of DVDD_GPU - C2039/C2038 to NC [PDN] Change Caps of DVDD_MODEM - C2502/C2534/C2523/C2522 to NC
2020.08.26	V03	File name	Modify the file name

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		SM6350_Control	
Size	Project		Rev
D	<Doc>		V1
Date:	Monday, October 18, 2021	Sheet 2 of	50



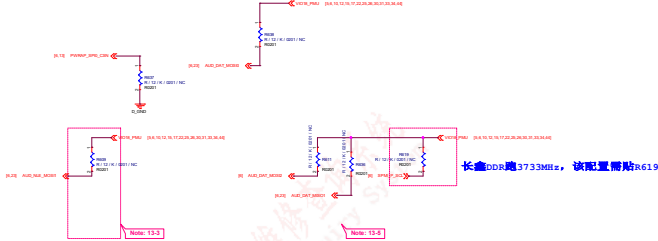
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 03.BB_POWER_PDN1			
Size D	Project	<Doc>	Rev vi
Date: Monday, October 18, 2021	Sheet 3	of 23	





Note 13-6: "AUD_DAT_MISO1" is trapping pin to select VEMC Voltage.

AUD_DAT_MISO1	VEMC Voltage
L (Default)	VEMC=3.0V for VCC of UFS2.1 device
H (by external PU)	VEMC=2.5V for VCC of UFS3.0 device



Schematic design notice:

Note 13-1: The load cap. have to be placed as close to REFP ball as possible.

Note 13-2: "PWRAP_SPI0_CSN" and "AUD_DAT_MISO1" pin features in trapping pin to enable JTAG.

PWRAP_SPI0_CSN	AUD_DAT_MISO1	AP_JTAG	ID_JTAG
H (Default)	L (Default)	N/A	N/A
(by external PU)	(by external PU)	SPHENT8	N/A
L (by external PD)	L	SPI0-ENT8	DMT16ENT11ENT18ENT19ENT20ENT21ENT22IDC7
L (by external PU)	H (by external PU)	MISC1_CLK MISC1_CMD MISC1_DATA MISC1_DAT1 MISC1_DAT2	N/A

Note 13-3: "AUD_NLE_MOSI1" features in trapping pin to enable JTAG over USB 2.0 port.

AUD_NLE_MOSI1	USB_JTAG
L (Default)	Disable serial JTAG over USB
H (by external PU)	DPIDM output JTAG

Note 13-4: "AUD_SYNC_MOSI" and "AUD_NLE_MOSI0" pin features in trapping pin to booting (eMMC/UFS/SPI NOR).

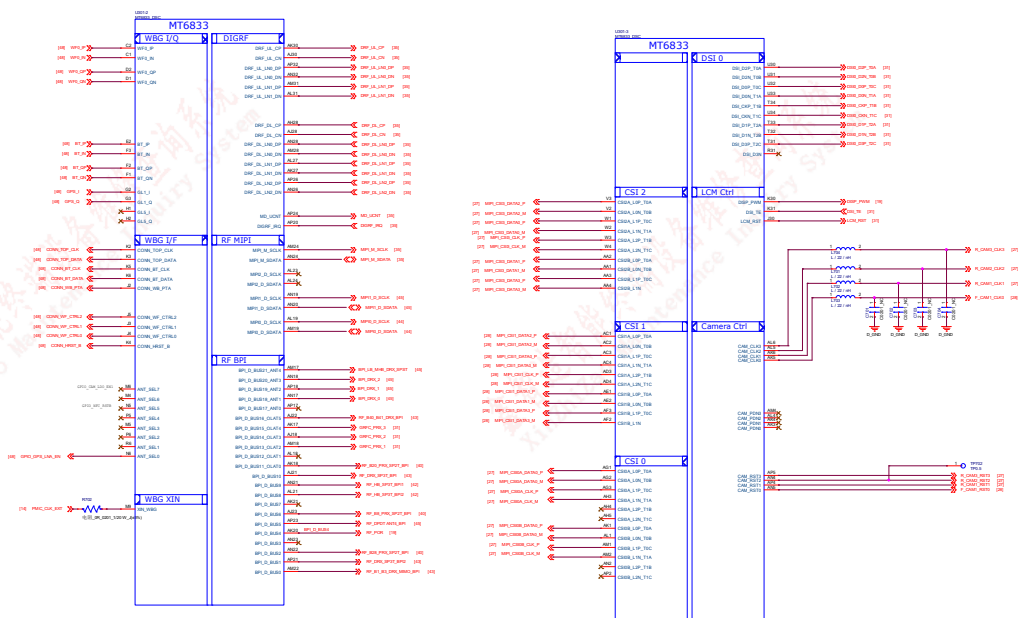
AUD_SYNC_MOSI	AUD_NLE_MOSI0	Storage Booting
L (Default)	L (Default)	Only UFS boot
L (by external PU)	H (by external PU)	Only eMMC boot
H (by external PU)	L	Reserved
H (by external PU)	H (by external PU)	Reserved

Note 13-5: "AUD_DAT_MOSI1" and "AUD_DAT_MOSI2" pin features in trapping pin to enable DDR.

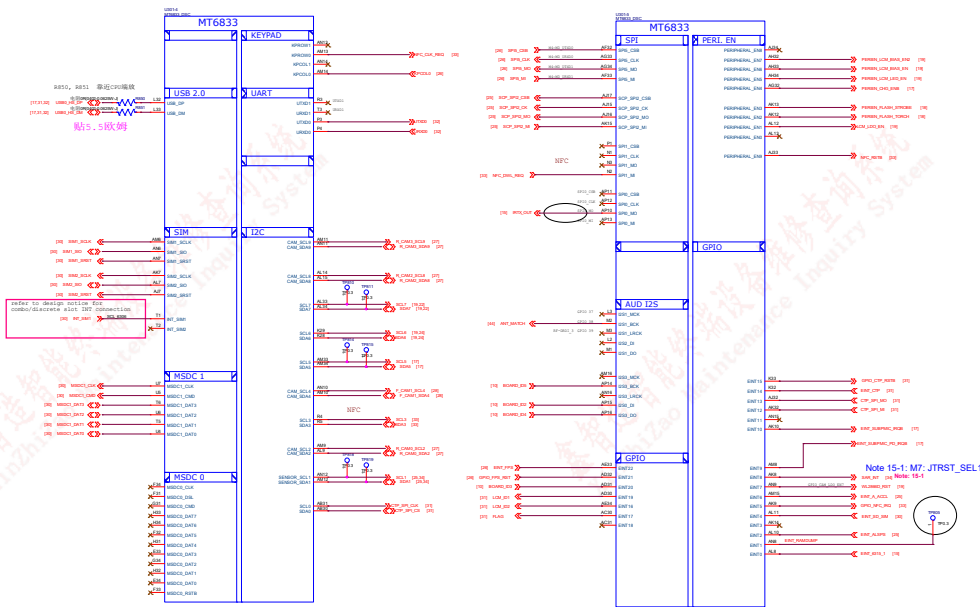
AUD_DAT_MOSI1	AUD_DAT_MOSI2	SPI0_P_SCL	DDR
L (Default)	L (Default)	L (Default)	LP4xDDR4-2400
H (by external PU)	H (by external PU)	L (Default)	LP4xDDR4-2400
H (by external PU)	L	L (Default)	Reserved
H (by external PU)	H (by external PU)	L (Default)	Reserved

Note: 1. SPI0_P_SCL for DDR 4200Mbps/3733Mbps Co-load, "L" = 1250Mbps, "H" (by external PU) = 3733Mbps

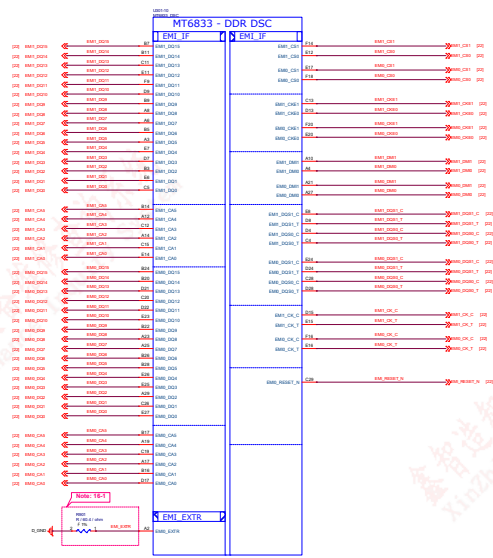
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		06.BB_1	
Size D	Project	<Doc>	Rev v1
Date:	Monday, October 18, 2021	Sheet 6 of 23	



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		07.BB_2	
Size	Project	<Doc>	Rev
D			v1
Date: Monday, October 18, 2021		Sheet 7 of 23	



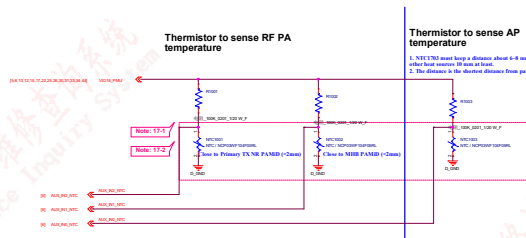
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		08.BB_3	
Size	Project	<Doc>	Rev
D			v1
Date:	Monday, October 18, 2021	Sheet 8 of 23	



Schematic design notice of "16_BB_4_interface" page:

Note 16-1: R4001 please select 60.4 ohm (1%) resistor

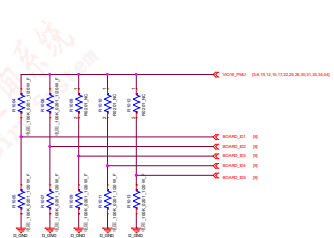
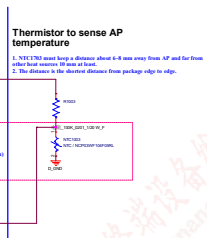
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		09.BB_4	
Size	Project	<Doc>	Rev
D			v1
Date:	Monday, October 18, 2021	Sheet 9 of 23	



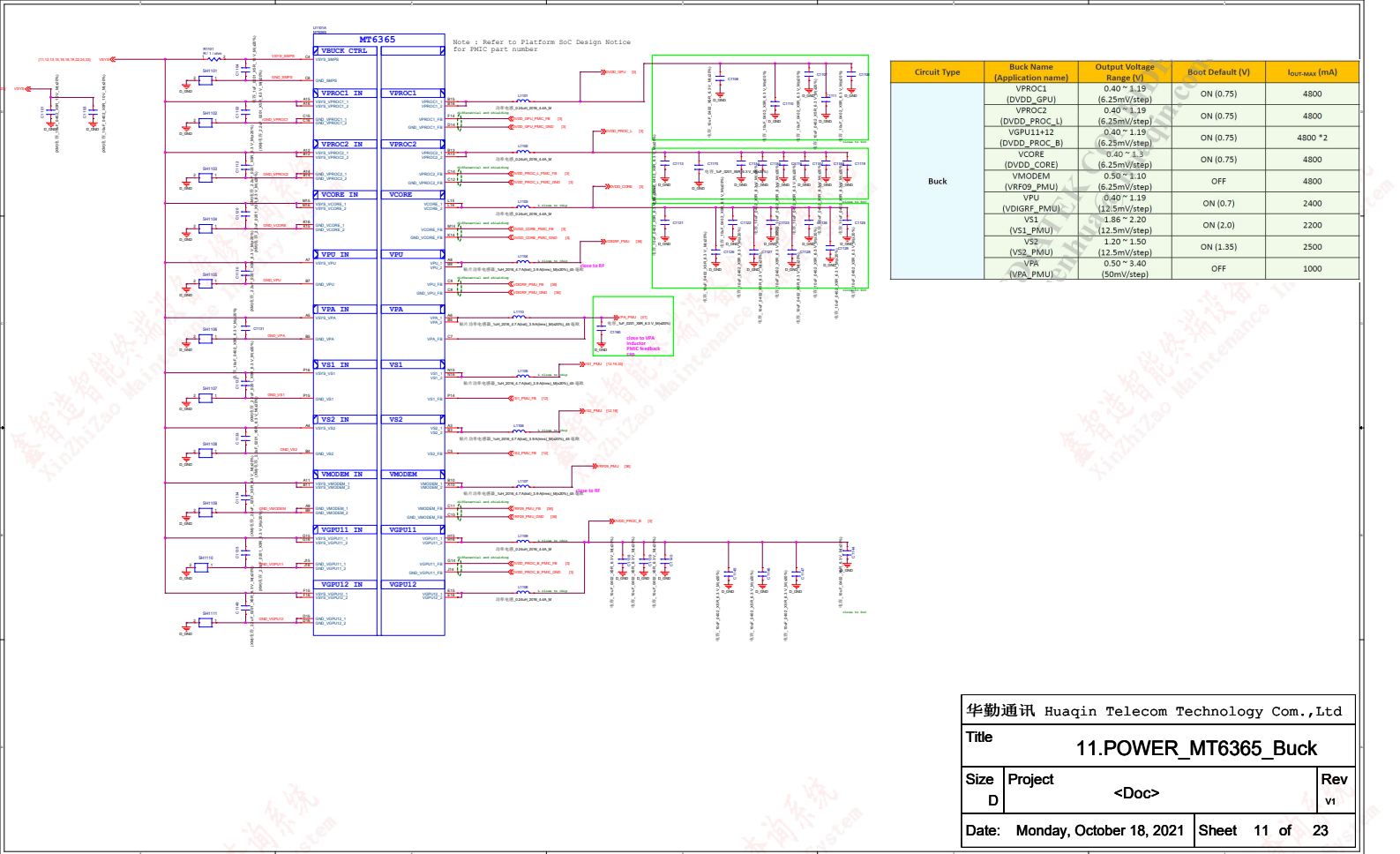
Schematic design notice:

Note 17-1: SW Default Configuration is as follows
 AUX_IN0_NTC is for AP Temp.
 AUX_IN1_NTC is for MHB PAMID Temp.
 AUX_IN2_NTC is for NR PAMID Temp.

Note 17-2: If your design or placement is different with SW Default Config.
 Please refer to "MTXXXX Thermal User Manual.doc" and modify SW setting.

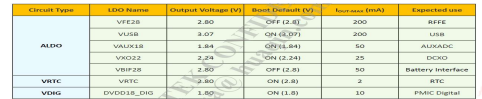


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		10.BB_AUXADC_Thermal/BOARD_ID	
Size	Project	<Doc>	Rev
D			V1
Date:	Monday, October 18, 2021	Sheet 10 of	23



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		11.POWER_MT6365_Buck	
Size	Project	<Doc>	Rev
D			v1
Date:	Monday, October 18, 2021	Sheet	11 of 23

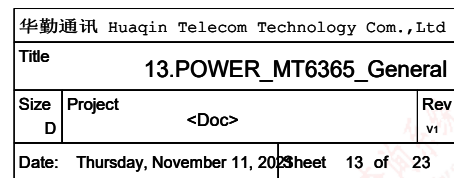
L71708 L71708		L71708	
MT6365		1. "Typical Cap" defined in design notice is the minimum cap. to LDO Cout.	
LDO IN	LDO	2. NC cap can move to application, if (PCB L<20nH, PCB R<0.2mohm)	
		=> value and placement of Cap, please refer design notice	

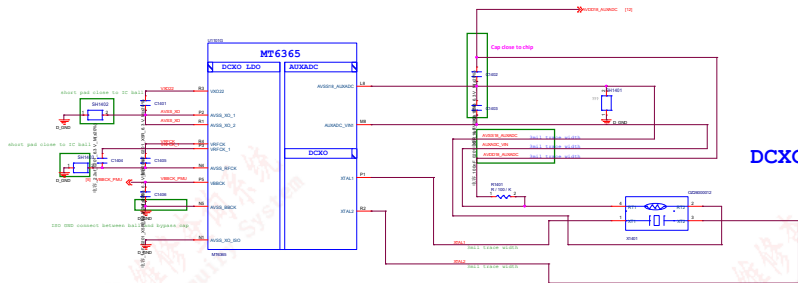


Camera Type	Video Name	Outward View (V)	Back Outward (B)	Surround (SR)	Expanded view
SD/DCI	VSD10	1.80	OFF (1.1E)	300	EFUSE
	VSD10B	1.80	ON (1.8E)	300	Audio
	VGAM0	1.80	OFF (1.1E)	300	Camera X
	VYSD2	1.80	ON (1.8E)	300	DRAMA
	VYF15	1.80	OFF (1.1E)	650	RF
	VSD2	1.80	ON (1.1E)	300	IO & Sensor
	VSD2B	1.80	OFF (1.1E)	1200	Communication
	VSD2C	1.80	ON (1.8E)	1200	uHARC / LPS
DCI/DCI HD	VSDC2	1.80 to 4.00 (1.8E)	ON (1.4E)	30	DCI/DCI
	VSDC2C	1.24	ON (1.24E)	30	DCI/DCI

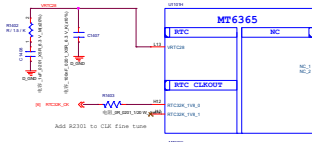
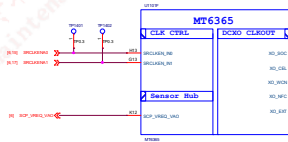
Genral Type	LDG Name	Output Voltage (V)	Rated Current (A)	Resonance (mH)	Engineered use
SLD02	V012	0.85	On (0.85)	300	DIGI4 SRAM (V01205_AEM4_PPAU)
	V0122	1.2	On (1.2)	300	AP Analog Module
	V01313	1.3	Off (1.3)	300	Connectivity
	VSRAM_PROC1	0.85 $\pm$ 1.3 (0.75 to V01205)	On (0.85)	600	CPUS SRAM
	VSRAM_PROC2	0.85 $\pm$ 1.3 (0.75 to V01205)	On (0.85)	600	CPUS SRAM
	VSRAM_OTHERS	0.85 $\pm$ 1.3 (0.75 to V01205)	On (0.75)	600	OTHERS SRAM
	VSRAM_AIO	0.85 $\pm$ 1.3 (0.75 to V01205)	On (0.85)	600	GPU SRAM
	V0122	1.2	On (1.2)	300	AP Analog Module

Date: Monday, October 18, 2021	Sheet 12 of 23
--------------------------------	----------------



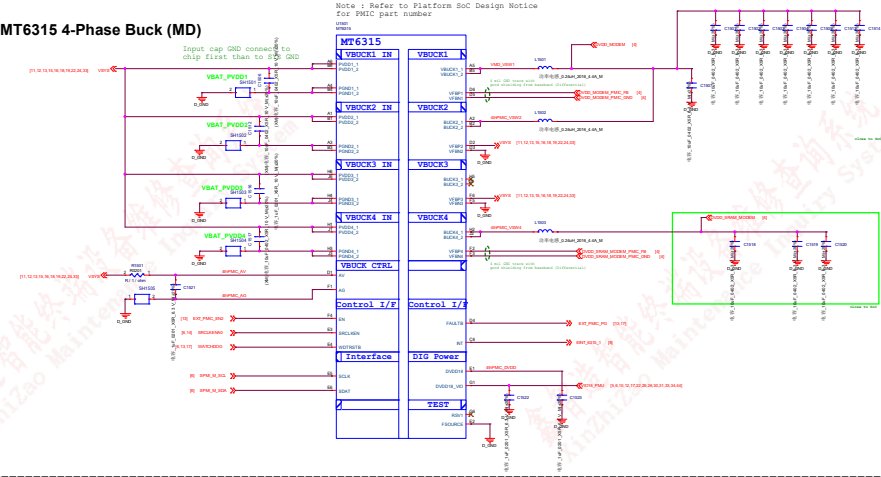


DCXO



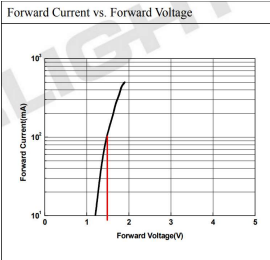
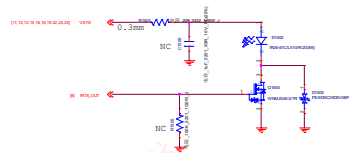
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		14.POWER_MT6365_Clock	
Size D	Project	<Doc>	Rev v1
Date:	Monday, October 18, 2021	Sheet 14 of	23

MT6315 4-Phase Buck (MD)



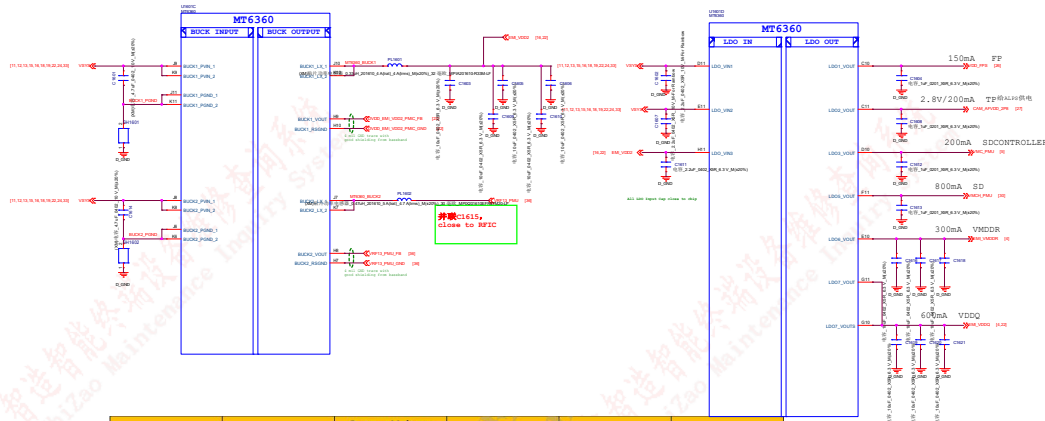
MT6315NP/B	Output Voltage Range [V]	Boot Default [V]	I _{OUT-MAx} [mA]	Expected use
BUCK1+2	0.40 ~ 1.19375 (6.25mV/step)	ON (0.825)	5000*2	DVDD_MODEM
BUCK3	0.40 ~ 1.19375 (6.25mV/step)	ON (0.75)	5000	---
BUCK4	0.40 ~ 1.19375 (6.25mV/step)	ON (0.825)	5000	DVDD_SRAM_MODEM

IR LED



华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 15.POWER_MT6315_VMD/IR_LED		
Size D	Project <Doc>	Rev v1
Date: Thursday, November 11, 2023		Sheet 15 of 23

Note : Refer to Platform SoC Design Notice for PMIC part number



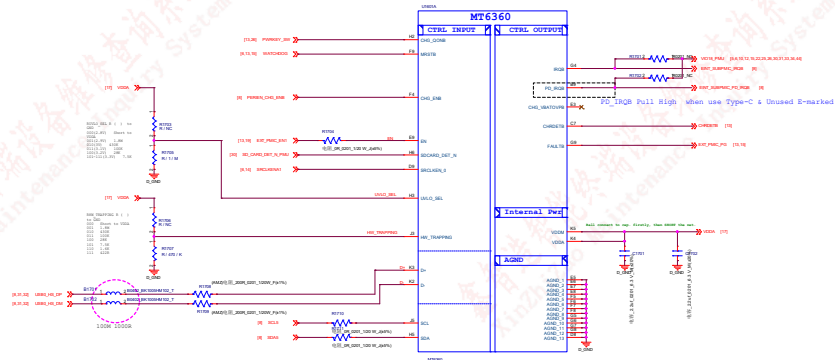
LDO1 VFP
LDO2 VTP
LDO3 VMC
LDO5 VMC
LDO6 VMC
LDO7 VMC

Circuit Type	Name	Output Voltage Range (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
Buck	BUCK1	0.3 ~ 1.3 (5mV/step)	ON (1.125)	3000	EMI_VDD2
	BUCK2	0.3 ~ 1.3 (5mV/step)	OFF (1.3)	3000	VRF13_PMU

Circuit Type	Name	Output Voltage (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
LDO	LDO1	1.8/2.0/2.1/2.5/2.7/ 2.8/2.9/3.0/3.1/3.3	OFF (1.8)	150	Fingerprint
	LDO2	1.8/2.0/2.1/2.5/2.7/ 2.8/2.9/3.0/3.1/3.3	OFF (1.8)	200	Touch Panel
	LDO3	1.8/2.9/2.8/3.0/3.3	OFF (3.0)	200	SD Card
	LDO5	2.9/3.0/3.3	OFF (3.0)	800	SD Card
	LDO6	0.75	ON (0.75)	300	EMI_VMDDR
	LDO7	0.6	ON (0.6)	600	EMI_VDDQ

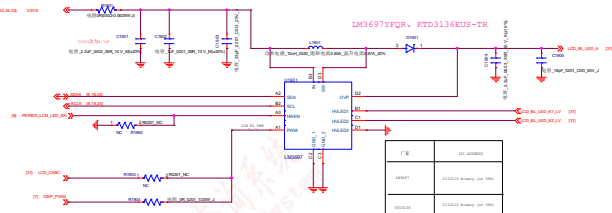
华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title 16.POWER_MT6360_BUCK_LDO*			
Size D	Project <Doc>	Rev V1	
Date: Monday, October 18, 2021	Sheet 16 of 23		

Note 1: Refer to Platform SoC Design Notice for PWC part number



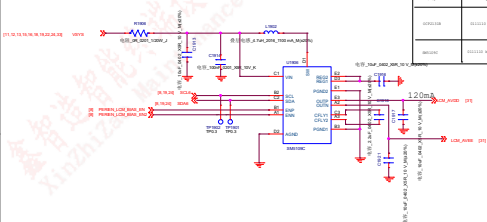
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 17.POWER_MT6360_General			
Size	Project	Rev	
D	<Doc>	v1	
Date:	Monday, October 18, 2021	Sheet	17 of 23

LCM Bias

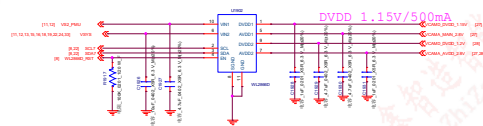


工廠	SO ADDRESS
UNION	SHANGHAI Winery (or SO)
SHANGHAI	SHANGHAI Winery (or SO)

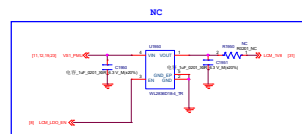
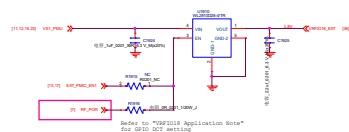
厂家	120 ADDRESS
SL28462	0x508
ST1804	0x508



厂家	xxx address
cpu类型	xxxxxx binary (see 30a)
总线	xxxxxx binary (see 30a)



External VIO18_LDO: option for power leakage from UART port on product line



华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title	19.POWER EXT
-------	--------------

Size	Project	<Doc>
D		

Rev	
v1	

Date: Thursday, November 11, 2023 Sheet 19 of 23

鑫智造智能终端设备维修查询系统
XinZhiZao Maintenance Inquiry System

鑫智造智能终端设备维修查询系统
XinZhiZao Maintenance Inquiry System

鑫智造智能终端设备维修查询系统
XinZhiZao Maintenance Inquiry System

鑫智造智能终端设备维修查询系统
XinZhiZao Maintenance Inquiry System

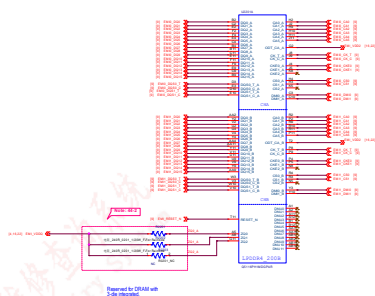
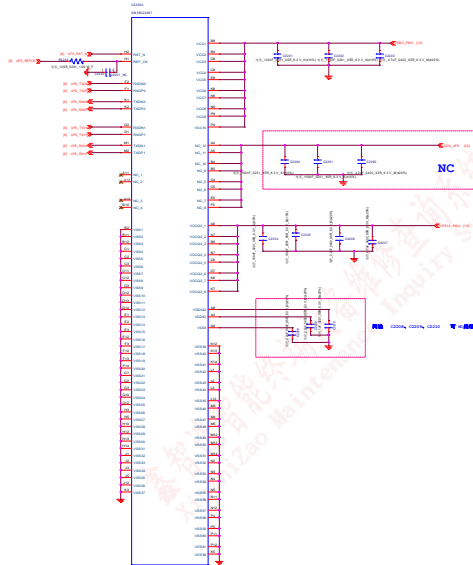
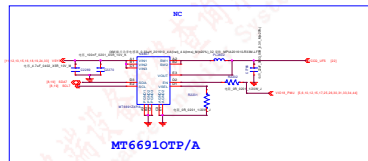
鑫智造智能终端设备维修查询系统
XinZhiZao Maintenance Inquiry System

鑫智造智能终端设备维修查询系统
XinZhiZao Maintenance Inquiry System

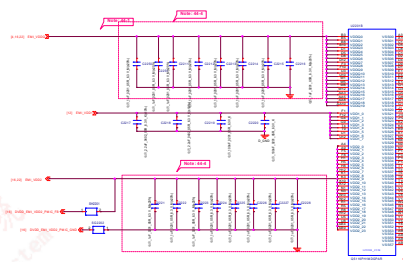
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		PM7250B_CHARGE/FG	
Size	Project	<Doc>	Rev
E			v1
Date:	Monday, October 18, 2021	Sheet	20 of 23



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		PM7250B_MISC/GPIO/GND	
Size	Project	<Doc>	Rev
D			v1
Date: Monday, October 18, 2021		Sheet 21 of	50

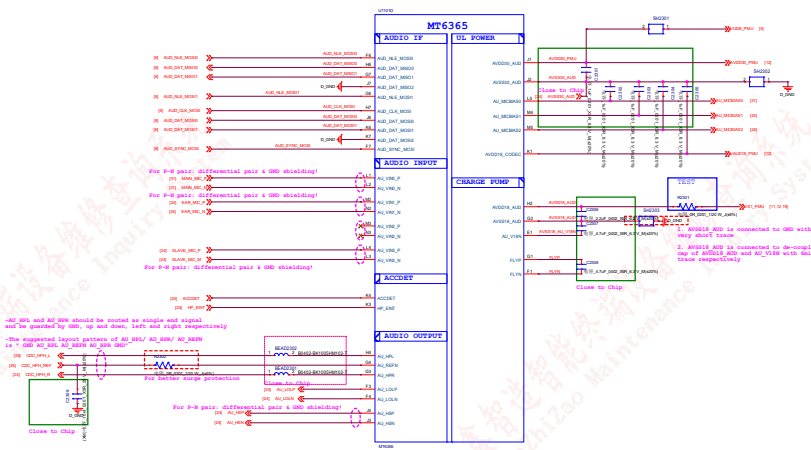


discrete LP4X
16Gb LPDDR4X



Schematic design notice of "44_Memory_4MMC_LPDDR4X"	
Note 44-1:	Please refer to power supply related page select: VDRAM2 / VDRAM1 output voltage properly for LPDDR4X.
Note 44-2:	DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ.
Note 44-3:	Please refer to eMCP vendor's datasheet or MTK common design notice to get the recommendation layout cap. value for VDDQ/VDDQ2/VDDQ3 power domains of 4MMC.
Note 44-4:	VDDQ VDDQ2 decoupling cap. (closed to DRAM bank). For other cap for PMIC >10uF, at PMIC page. Please also refer to MMD and layout guide for placement.

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 01_Block_Diagram		
Size E	Project J19_MB_V1	Rev v1
Date: Monday, October 18, 2021	Sheet 22 of 48	



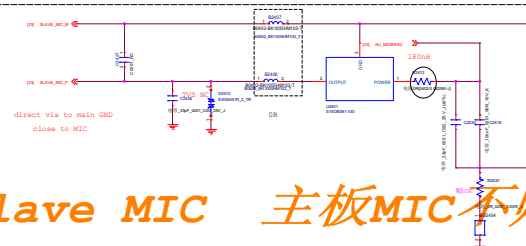
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		23.PERI_AUDIO_PMIC-IF	
Size	Project	<Doc>	Rev
D			v1
Date:	Monday, October 18, 2021	Sheet	23 of 23

国内	D2406, D2407, D2408 (NC), D2409
国际	D2401, D2402, D2403 (NC), D2404



Note 4-4: 1. Need do SW configuration for DCC mode
2. Analog input impedance is as below,
please customer check the detail requirement
with mic vendor if the mic used has special
requirement on the input impedance of analog input

REF257	UL Mode	MIN	TYP	MAX	Unit
C	000	-	-	10	E

[illegible]

Slave MIC 主板MIC不贴

华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title	24.MIC/RCV/SPK/Earphone
-------	-------------------------

Size	Project	Rev
D	<Doc>	v1

Date: Monday, December 06, 2021 Sheet 24 of 50

M-Sensor

厂号	001 000000
物料号	0ch
物料描述	2ch

A sensor

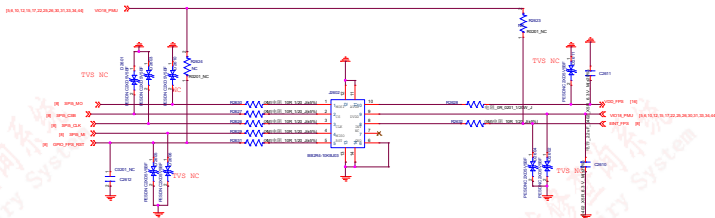
default SPI (if sensor hub support)

厂号	001 000000
物料号	000000
物料描述	000000

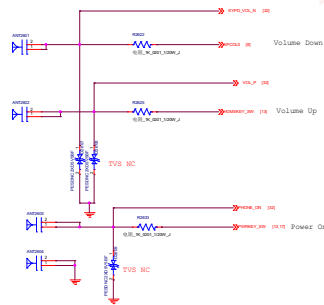
ALS+PS

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		25.SENSOR	
Size	Project	<Doc>	Rev
D			V1
Date:	Monday, October 18, 2021	Sheet	25 of 50

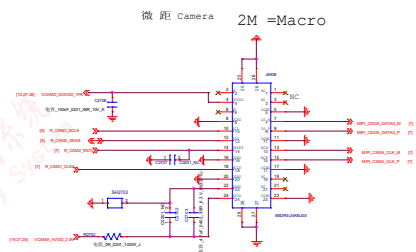
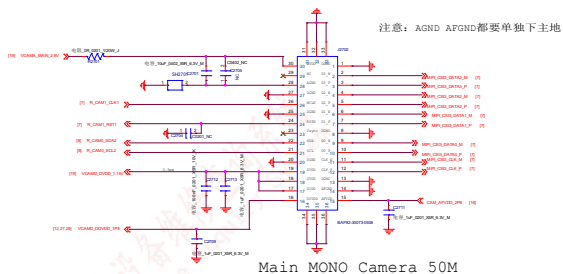
Finger Print



KEY BTB

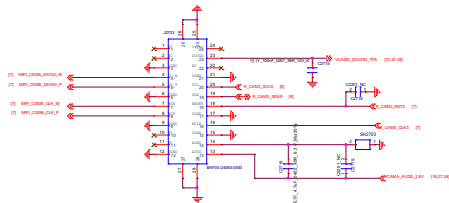


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 26.FP/KEY			
Size D	Project <Doc>	Rev v1	
Date: Tuesday, November 02, 202	Sheet 26 of	50	

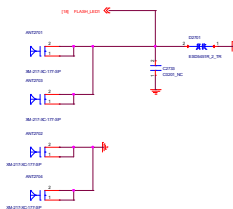


微距不贴

景深 Camera 2M=Depth



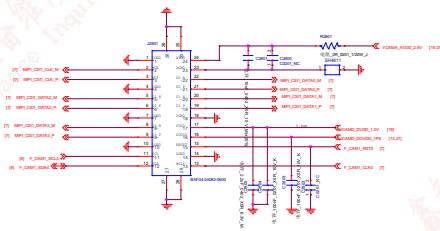
Flash LED 5V Boost



Flash 下方地要完整且足够大, 散热一定要好

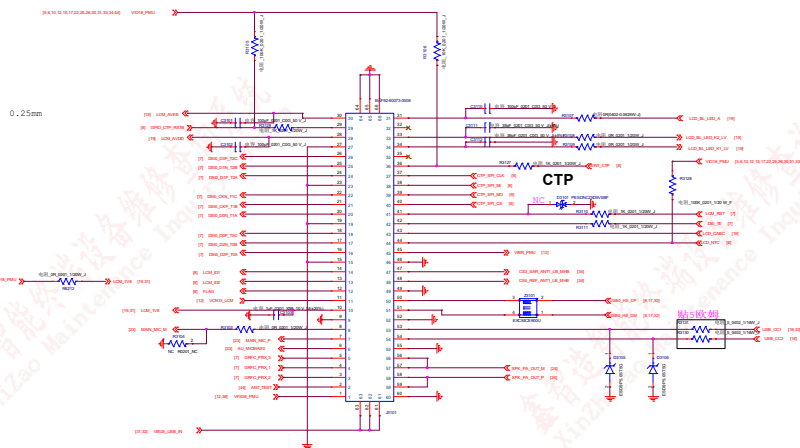
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		27.REAR_CAM	
Size	Project	<Doc>	Rev
D			V1
Date:	Monday, October 18, 2021	Sheet	27 of 50

Front Camera - 8M

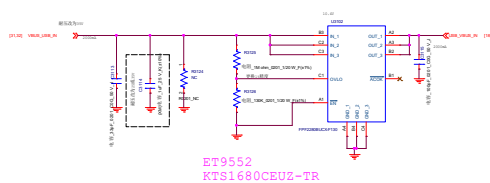


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		28.FRONT_CAM	
Size	Project	<Doc>	Rev
D			V1
Date:	Monday, October 18, 2021	Sheet	28 of 50

LINK



OVP



华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title		31.60PIN_LCD_LINK/OVP	
Size	Project	<Doc>	Rev
D			V1
Date:	Friday, December 24, 2021	Sheet	31 of 50

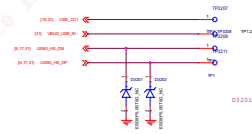
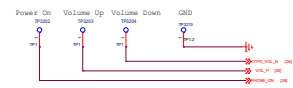
VBAT TEST POINT



VBAT

KEYPAD

KEY BTB

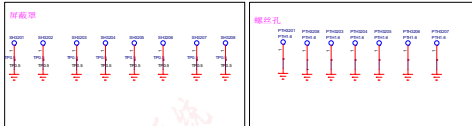


VCHG

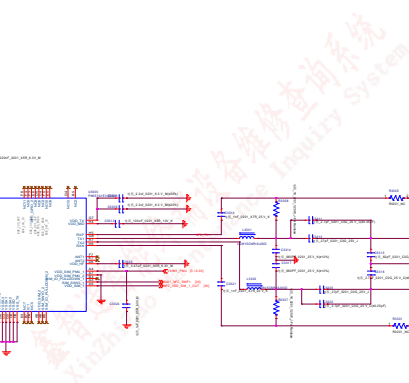
UART TP



JTAG

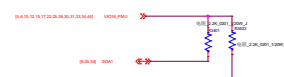
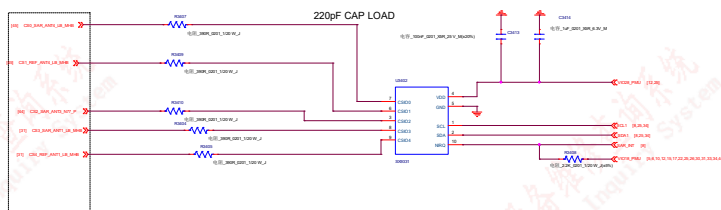


华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 32.TEST_POINT/SHIELD		
Size D	Project <Doc>	Rev V1
Date: Monday, October 18, 2021	Sheet 32 of 50	



知道智能终端设备维修查询系统
XinZhiZao Maintenance Inquiry System

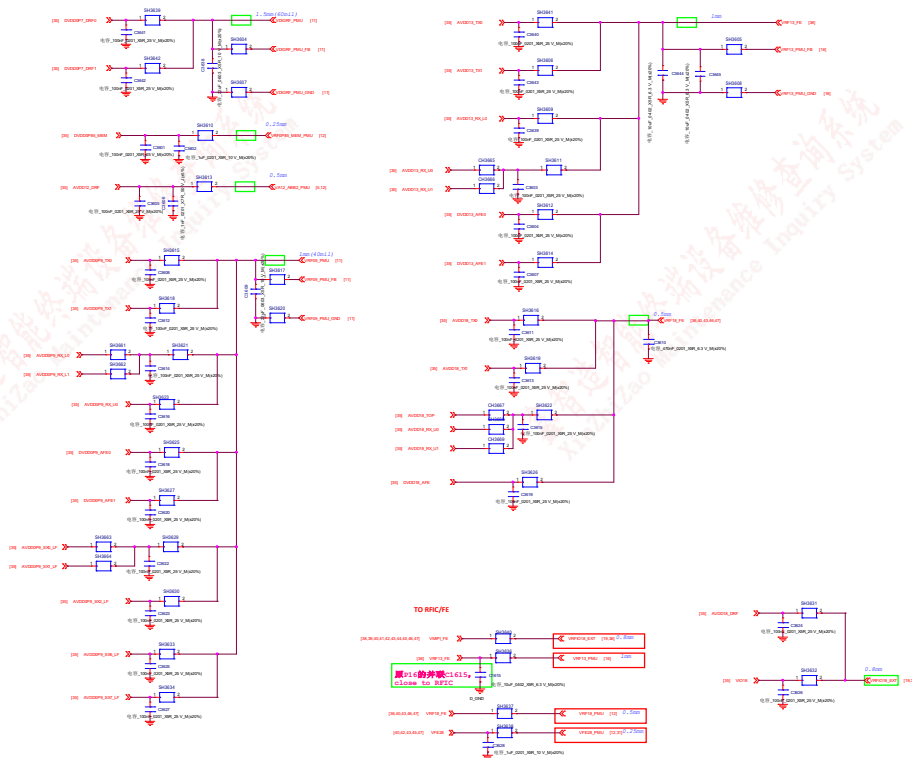
SAR_sensor



- Design notice:
- 1) CS0_1 nets are differential pairs;
 - 2) The width for each line, should be less than 0.1mm;
 - 3) 0.2mm should be guaranteed between the ground and differential pairs;
 - 4) And also, the distance between the differential line should be more than 0.2mm;
 - 5) All the CSx nets should be protected by surrounding GND;

华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title		SAR_sensor	
Size	Project	<Doc>	Rev
D			V1
Date:	Friday, November 12, 2021	Sheet 34 of	50

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title			
MT6190M_PWR			
Size D	Project <Doc>	Rev v1	
Date: Monday, October 18, 2021		Sheet 36 of 50	

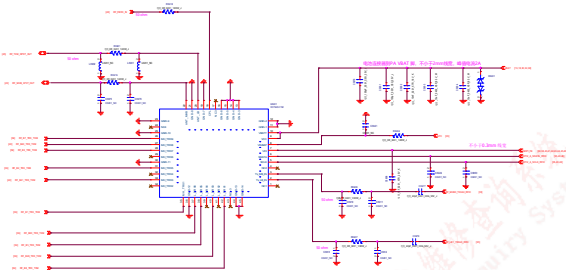




华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title NC_FBRX		
Size E	Project <Doc>	Rev v1
Date: Monday, October 18, 2021	Sheet 37 of	50

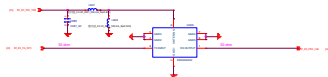
MAIN_ANT TXM_GSM

LB
MHB

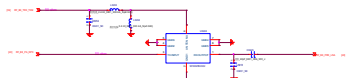


TRX_LB

B5 TX



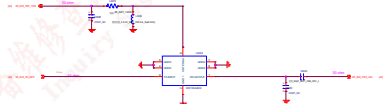
B8 TX



B20 TX

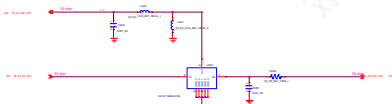


B28 TX

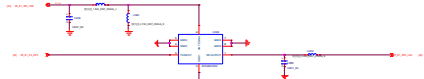


TRX_MHB

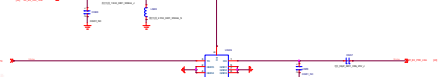
B2 TX



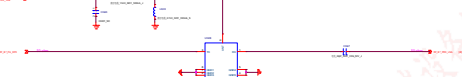
B1 TX



B3 TX



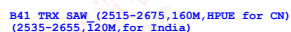
B7 TX



B34 /B39 PRX

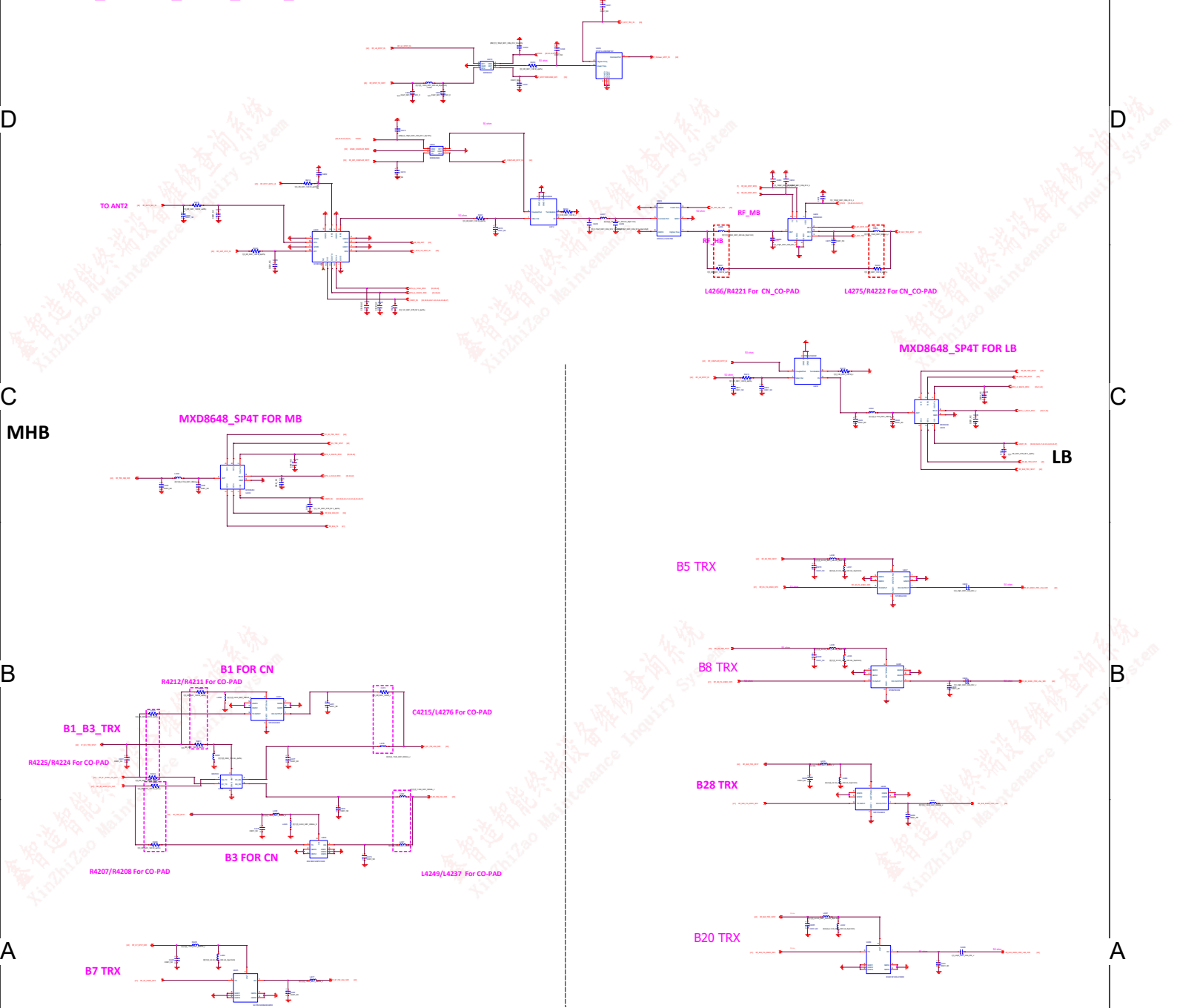


华勤通讯 Huaqin Telecom Technology Co.,Ltd		
Title TX_Module		
Size	Project Custom	Rev v1
Date: Friday, November 12, 2021		Sheet 39 of 50



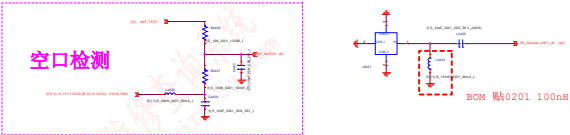
华勤通讯 Huaqin Telecom Technology Co., Ltd Title 4/5G_MHB_PA_Phase5N		
Size Custom	Project <Doc>	Rev /v1
Date: Monday, October 18, 2021	Sheet	40 of 50

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title MMBB PA_MB ENDC			
Size E	Project <Doc>		Rev v1
Date: Friday, November 12, 2021		Sheet 41 of 50	

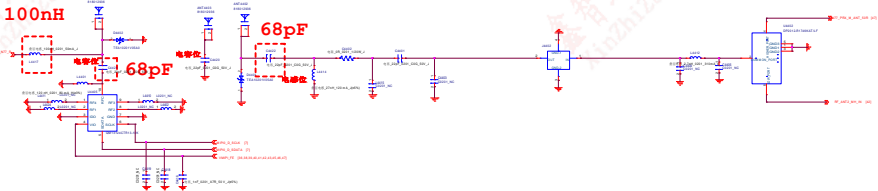


Title		
MMMB PA_VC7643_ENDC_MHB_TRX		
Size	Project	Rev
Custom	<Doc>	v1
Date: Friday, November 12, 2021		Sheet 42 of 50

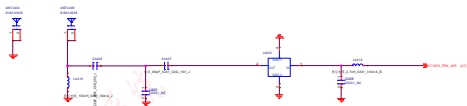
ANT1 (LB_MHB_P) to 小板



ANT2 (MHB_PM/N77_PRX_MIMO)

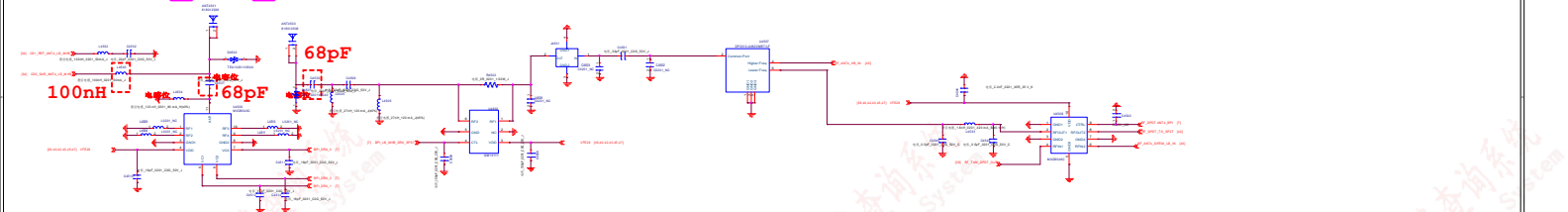


ANT3 (N77_PRX)

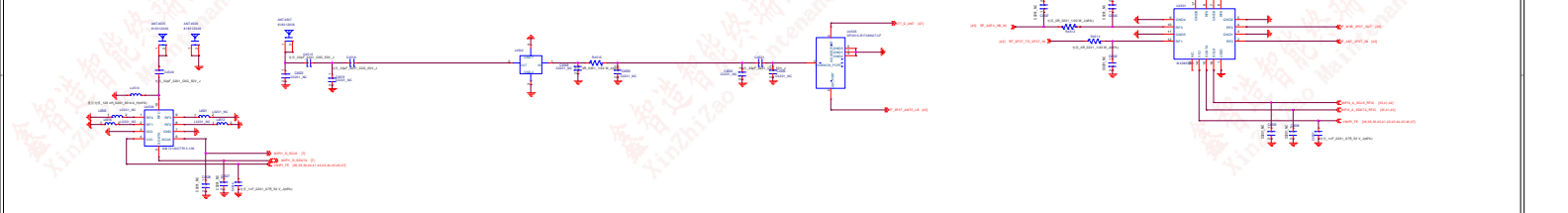


华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title ANT_P_PM		
Size E	Project <Doc>	Rev v1
Date: Monday, October 18, 2021		Sheet 44 of 50

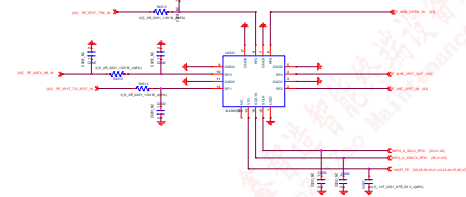
ANT4 (LB_MHB_D)



ANT5 (MHB_DM/N77_D)



AsDIV_3P3T



接地弹片

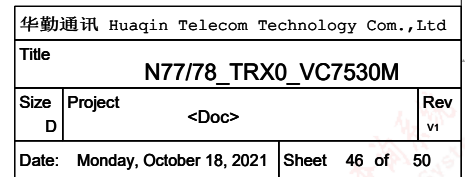


ANT6 (N77_DM_HB_DRX) to 小板



华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title		ANT_D_DM	
Size	Project	<Doc>	Rev
E			v1
Date:	Friday, October 22, 2021	Sheet	45 of 50

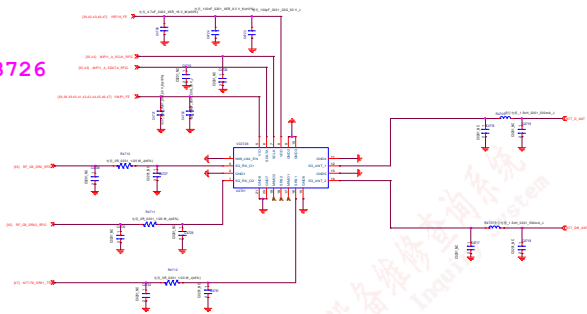
NTC温度检测
见P10的R1002 and NTC1002



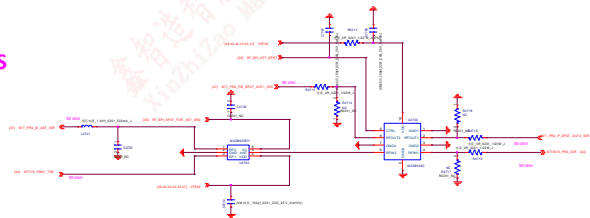
VC3726

N77/78 DRX

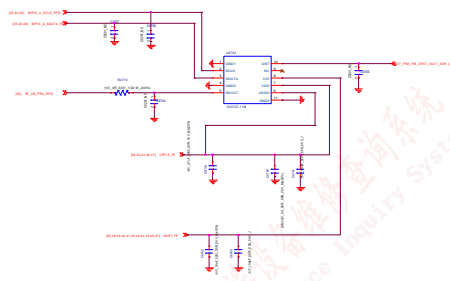
N77/78 DRX MIMO



DPDT+SP2T FOR N77/78 SRS



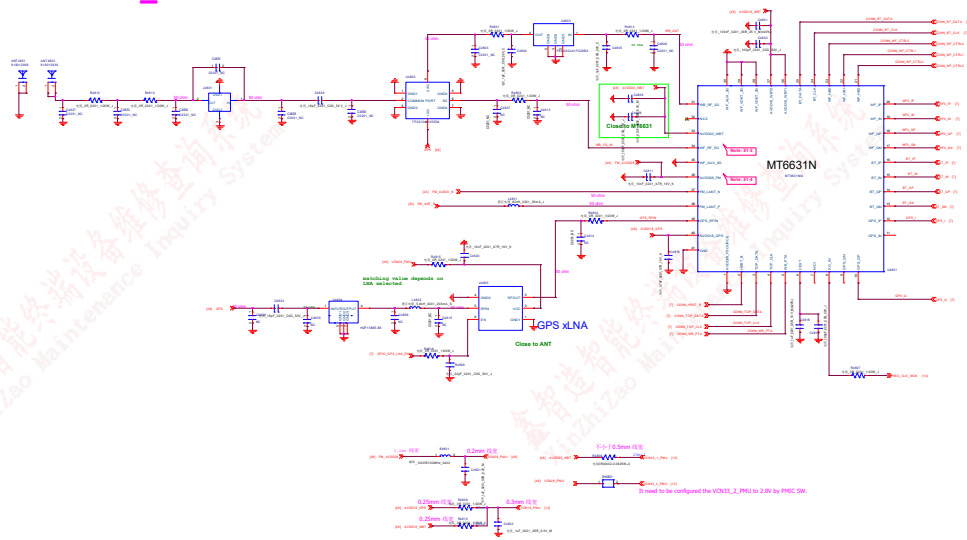
VC3727-11M_N77/78 PRX MIMO



华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title VC3726		
Size E	Project <Doc>	Rev v1
Date: Saturday, November 13, 2021	Sheet 47 of 50	

ANT7 WiFi 2.4G & 5G_GPS

MT6631

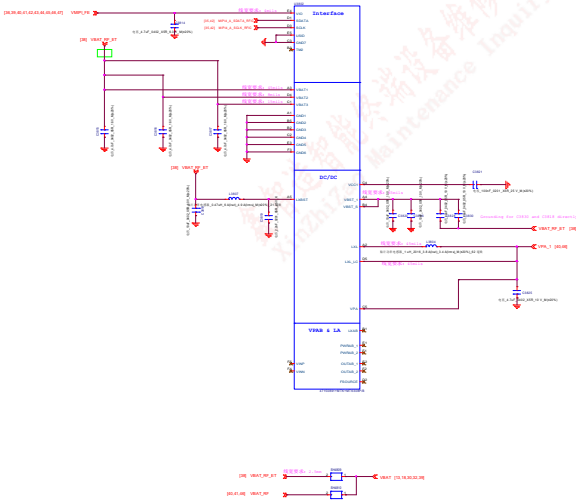


Schematic design notice of "61_CONNECTIVITY_CONSYS_MT6631"

- Note 51-1: For P6015 case, please select Q402 size or larger one.
- Note 51-2: Please refer to MT6631 Baseband design index for VCN33_L2D selection guide.
- Note 51-3: If WPI 55 not support, contact pin 34(WPI_RP_35) to GND.
- Note 51-4: Pin 36 (400000_FIM) must be connected to VCN33_L2_P30 to 2.0V by PMIC SW.

华勤通讯 Huaqin Telecom Technology Com., Ltd		
Title MT6631		
Size E	Project <Doc>	Rev v1
Date: Monday, October 18, 2021 Sheet 46 of 50		

MT6308 FOR Phase 5N PA_N77_PA



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		RF_APT_MT6308M	
Size	Project	<Doc>	Rev
E			v1
Date: Monday, October 18, 2021		Sheet 50 of 50	